W4 - Data Cleaning

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2026-02-02

1) Project topic and dataset description

This project examines longitudinal trends in U.S. suicide mortality rates and how rates differ by demographic group (sex, race/ethnicity, age) using national-level rate estimates.

Each row represents one national estimate of the suicide death rate per 100,000 resident population for a specific year, age group, and demographic subgroup defined by the stratification fields (`STUB_NAME` + `STUB_LABEL`). The `FLAG` field indicates N/A (ellipses) or unreliable estimates (asterisk, often due to small counts).

2) Read data

```
library(readr)
library(dplyr)
library(tidyr)
library(ggplot2)
library(knitr)
library(stringr)

file_path <- "/Users/user/Downloads/Death_rates_for_suicide__by_sex__race__Hispanic_origin__and_age__Uni
# If you prefer not to hardcode paths, comment above and use:
# file_path <- file.choose()

stopifnot(file.exists(file_path))

suicide_raw <- read_csv(file_path, show_col_types = FALSE)

# small, safe previews (do NOT print whole data)
dim(suicide_raw)
```

```
## [1] 6390   13
```

```
head(suicide_raw, 8)
```

```
## # A tibble: 8 x 13
##   INDICATOR     UNIT  UNIT_NUM STUB_NAME STUB_NAME_NUM STUB_LABEL STUB_LABEL_NUM
##   <chr>         <chr>    <dbl> <chr>             <dbl> <chr>               <dbl>
## 1 Death rates ~ Deat~        1 Total                 0 All perso~              0
## 2 Death rates ~ Deat~        1 Total                 0 All perso~              0
```

```
## 3 Death rates ~ Deat~        1 Total                 0 All perso~              0
## 4 Death rates ~ Deat~        1 Total                 0 All perso~              0
## 5 Death rates ~ Deat~        1 Total                 0 All perso~              0
## 6 Death rates ~ Deat~        1 Total                 0 All perso~              0
## 7 Death rates ~ Deat~        1 Total                 0 All perso~              0
## 8 Death rates ~ Deat~        1 Total                 0 All perso~              0
## # i 6 more variables: YEAR <dbl>, YEAR_NUM <dbl>, AGE <chr>, AGE_NUM <dbl>,
## #   ESTIMATE <dbl>, FLAG <chr>
```

3) Prioritize variables for research (minimum 10 variables)

Your questions focus on: - Time: YEAR - Age groups: AGE (categorical) + AGE_NUM (ordering code) - Demographic stratification: STUB_NAME / STUB_LABEL (and their numeric codes) - Outcome: ESTIMATE (rate) - Data quality: FLAG

To meet the requirement (at least 10 variables; at least 6 numeric; at least2 categorical), I will keep exactly 10 variables: - Numeric (6): UNIT_NUM, STUB_NAME_NUM, STUB_LABEL_NUM, YEAR, AGE_NUM, ESTIMATE - Categorical (4): STUB_NAME, STUB_LABEL, AGE, FLAG

Before cleaning, I also check unit consistency and basic structure.

```r
# column types
type_table <- tibble(
  column = names(suicide_raw),
  class  = sapply(suicide_raw, function(x) paste(class(x), collapse = ", "))
)
kable(type_table, caption = "Column types (raw data)")
```

Table 1: Column types (raw data)

column	class
INDICATOR	character
UNIT	character
UNIT_NUM	numeric
STUB_NAME	character
STUB_NAME_NUM	numeric
STUB_LABEL	character
STUB_LABEL_NUM	numeric
YEAR	numeric
YEAR_NUM	numeric
AGE	character
AGE_NUM	numeric
ESTIMATE	numeric
FLAG	character

```r
# key data quality checks
data.frame(
  total_rows = nrow(suicide_raw),
  duplicate_rows = sum(duplicated(suicide_raw)),
  missing_estimate = sum(is.na(suicide_raw$ESTIMATE)),
  missing_flag = sum(is.na(suicide_raw$FLAG))
) %>% kable(caption = "Quality checks (raw data)")
```

Table 2: Quality checks (raw data)

total_rows	duplicate_rows	missing_estimate	missing_flag
6390	0	906	5484

4) Clean your data

4.1 Units: prevent mixing incompatible measurements

Some datasets include multiple units (e.g., crude vs age-adjusted). I will choose one unit and filter to it.

```
unit_values <- sort(unique(suicide_raw$UNIT))
unit_values
```

```
## [1] "Deaths per 100,000 resident population, age-adjusted"
## [2] "Deaths per 100,000 resident population, crude"
```

```
# Prefer an age-adjusted "Deaths per 100,000" unit if present; otherwise keep the first unit.
unit_to_keep <- unit_values[
  which(str_detect(tolower(unit_values), "death") &
        str_detect(tolower(unit_values), "100,000"))
][1]

if (is.na(unit_to_keep)) {
  unit_to_keep <- unit_values[1]
}

unit_to_keep
```

```
## [1] "Deaths per 100,000 resident population, age-adjusted"
```

4.2 YEAR vs YEAR_NUM redundancy

If YEAR and YEAR_NUM match exactly, YEAR_NUM is redundant.

```
year_matches <- all(suicide_raw$YEAR == suicide_raw$YEAR_NUM, na.rm = TRUE)
year_matches
```

```
## [1] FALSE
```

4.3 Apply cleaning rules

Cleaning actions: 1. Filter to one unit (UNIT == unit_to_keep) 2. Remove exact duplicates (distinct()) 3. Remove rows where ESTIMATE is missing (cannot analyze missing outcomes) 4. Remove N/A rows indicated by FLAG == "..." (not applicable) 5. Do NOT auto-remove "outliers" for YEAR or ESTIMATE (extremes can be real; YEAR is not an outlier concept) 6. Drop YEAR_NUM if redundant 7. Keep the prioritized 10 variables (6 numeric + 4 categorical) (NOTE: understand that some of the variables are a bit repetitve - if this is an issue please communicate this with me via EMAIL and let me know if a different dataset is needed.)

```r
suicide_clean <- suicide_raw %>%
  filter(UNIT == unit_to_keep) %>%
  distinct() %>%
  filter(!is.na(ESTIMATE)) %>%
  filter(is.na(FLAG) | FLAG != "...") %>%
  { if (year_matches) select(., -YEAR_NUM) else . } %>%
  select(
    # categorical (4)
    STUB_NAME, STUB_LABEL, AGE, FLAG,
    # numeric (6)
    UNIT_NUM, STUB_NAME_NUM, STUB_LABEL_NUM, YEAR, AGE_NUM, ESTIMATE
  )

dim(suicide_clean)
```

```
## [1] 723  10
```

```r
head(suicide_clean, 8)
```

```
## # A tibble: 8 x 10
##   STUB_NAME STUB_LABEL  AGE    FLAG  UNIT_NUM STUB_NAME_NUM STUB_LABEL_NUM  YEAR
##   <chr>     <chr>       <chr>  <chr>    <dbl>         <dbl>          <dbl> <dbl>
## 1 Total     All persons All a~ <NA>         1             0              0  1950
## 2 Total     All persons All a~ <NA>         1             0              0  1960
## 3 Total     All persons All a~ <NA>         1             0              0  1970
## 4 Total     All persons All a~ <NA>         1             0              0  1980
## 5 Total     All persons All a~ <NA>         1             0              0  1981
## 6 Total     All persons All a~ <NA>         1             0              0  1982
## 7 Total     All persons All a~ <NA>         1             0              0  1983
## 8 Total     All persons All a~ <NA>         1             0              0  1984
## # i 2 more variables: AGE_NUM <dbl>, ESTIMATE <dbl>
```

4.4 Confirm we meet the "minimum variables" rule and types

```r
data.frame(
  n_variables = ncol(suicide_clean),
  n_numeric = sum(sapply(suicide_clean, is.numeric)),
  n_categorical = sum(sapply(suicide_clean, function(x) is.character(x) || is.factor(x)))
) %>% kable(caption = "Variable count check (cleaned data)")
```

Table 3: Variable count check (cleaned data)

n_variables	n_numeric	n_categorical
10	6	4

```r
kable(
  tibble(variable = names(suicide_clean),
         class = sapply(suicide_clean, function(x) paste(class(x), collapse = ", "))),
  caption = "Column types (cleaned data)"
)
```

Table 4: Column types (cleaned data)

variable	class
STUB_NAME	character
STUB_LABEL	character
AGE	character
FLAG	character
UNIT_NUM	numeric
STUB_NAME_NUM	numeric
STUB_LABEL_NUM	numeric
YEAR	numeric
AGE_NUM	numeric
ESTIMATE	numeric

5) Summary statistics (cleaned data)

5.1 Numeric summary table (at least 4 numeric variables)

ESTIMATE is the main outcome (rate). YEAR is time. AGE_NUM and STUB_LABEL_NUM are numeric ordering/coding variables that support sorting and subgroup structure.

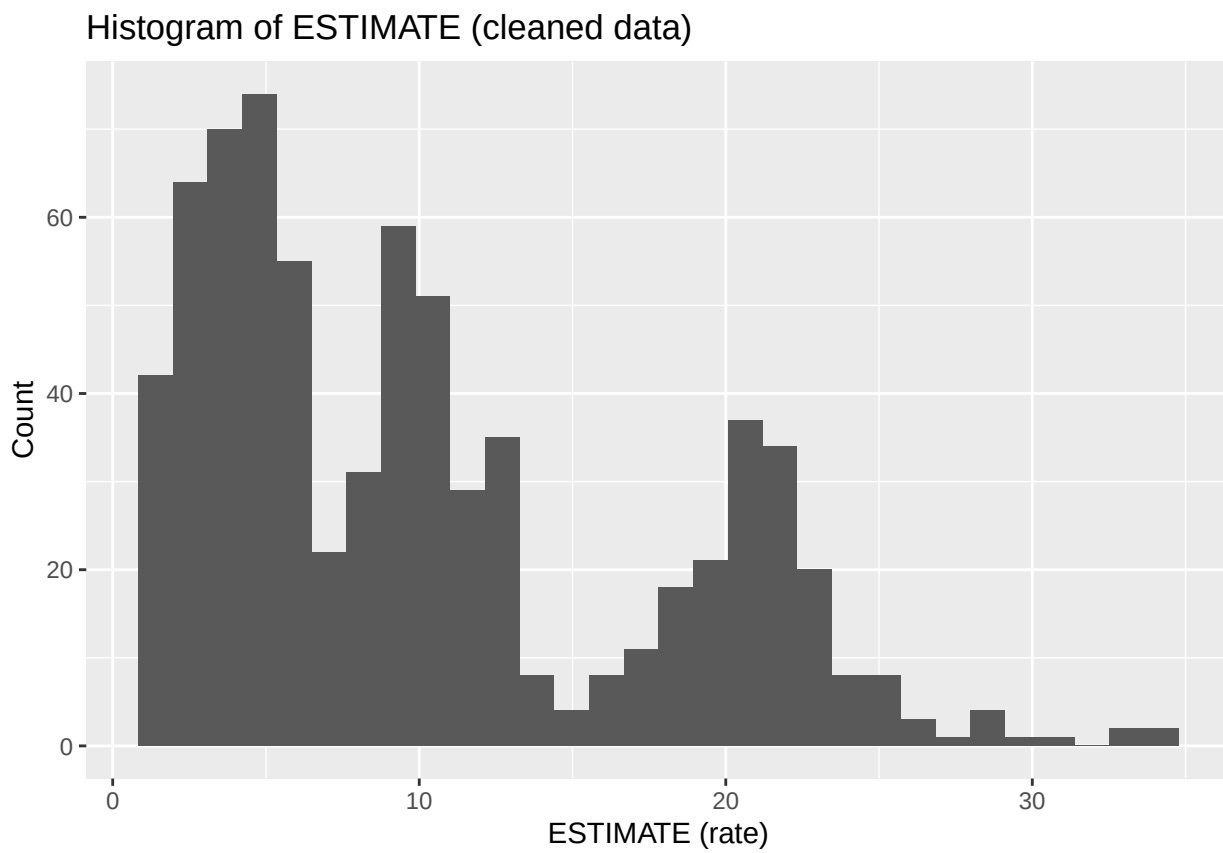
```
num_vars <- c("ESTIMATE", "YEAR", "AGE_NUM", "STUB_LABEL_NUM")

numeric_summary <- suicide_clean %>%
  summarise(across(all_of(num_vars),
                   list(
                     min  = ~ min(., na.rm = TRUE),
                     max  = ~ max(., na.rm = TRUE),
                     mean = ~ mean(., na.rm = TRUE),
                     sd   = ~ sd(., na.rm = TRUE)
                   ),
                   .names = "{.col}__{.fn}")) %>%
  pivot_longer(everything(), names_to = "metric", values_to = "value") %>%
  separate(metric, into = c("variable","stat"), sep = "__") %>%
  pivot_wider(names_from = stat, values_from = value) %>%
  select(variable, min, max, mean, sd)

kable(numeric_summary, digits = 3, caption = "Numeric summary statistics (cleaned data)")
```

Table 5: Numeric summary statistics (cleaned data)

variable	min	max	mean	sd
ESTIMATE	1.4	34.20	10.271	7.287
YEAR	1950.0	2018.00	2000.361	12.958
AGE_NUM	0.0	0.00	0.000	0.000
STUB_LABEL_NUM	0.0	6.26	4.437	1.705

5.2 Simple plot (histogram)

```
ggplot(suicide_clean, aes(x = ESTIMATE)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Histogram of ESTIMATE (cleaned data)",
    x = "ESTIMATE (rate)",
    y = "Count"
  )
```



5.3 Categorical summaries (at least 2 categorical variables)

For each categorical variable, I report: - number of unique values - frequency table (top levels only as the rest aren't as significant)

```
n_unique_age <- n_distinct(suicide_clean$AGE)
n_unique_age
```

Categorical variable 1: AGE

```
## [1] 1
```

```r
age_freq <- suicide_clean %>%
  count(AGE, sort = TRUE)

kable(head(age_freq, 15), caption = "AGE frequency table (top 15)")
```

Table 6: AGE frequency table (top 15)

AGE	n
All ages	723

```r
n_unique_stub <- n_distinct(suicide_clean$STUB_LABEL)
n_unique_stub
```

Categorical variable 2: STUB_LABEL

```
## [1] 24
```

```r
stub_freq <- suicide_clean %>%
  count(STUB_LABEL, sort = TRUE)

kable(head(stub_freq, 15), caption = "STUB_LABEL frequency table (top 15)")
```

Table 7: STUB_LABEL frequency table (top 15)

STUB_LABEL	n
Female: Black or African American	43
Female: White	43
Male: Black or African American	43
Male: White	43
All persons	42
Female	42
Male	42
Female: American Indian or Alaska Native	40
Male: American Indian or Alaska Native	40
Female: Asian or Pacific Islander	39
Male: Asian or Pacific Islander	39
Female: Hispanic or Latino: All races	35
Female: Not Hispanic or Latino: White	35
Male: Hispanic or Latino: All races	35
Male: Not Hispanic or Latino: White	35

(Additional categorical checks, not required but useful)

```r
n_unique_stubname <- n_distinct(suicide_clean$STUB_NAME)
n_unique_stubname
```

```
## [1] 6
```

```r
stubname_freq <- suicide_clean %>%
  count(STUB_NAME, sort = TRUE)

kable(head(stubname_freq, 15), caption = "STUB_NAME frequency table (top 15)")
```

Table 8: STUB_NAME frequency table (top 15)

STUB_NAME	n
Sex and race	324
Sex and race and Hispanic origin	256
Sex	84
Total	42
Sex and race and Hispanic origin (Single race)	11
Sex and race (Single race)	6

6) Research questions

These questions come from my initial proposal (and are still appropriate after cleaning). I refined wording slightly to match the cleaned structure (YEAR, AGE, STUB fields, ESTIMATE).

1. Disparity: How do suicide death rates differ by sex and race/ethnicity, and have these disparities widened or narrowed over time?
2. Age: Which age groups are the most affected by suicide, and have the highest-risk age groups changed over time?
3. Sex: Are men or women more impacted by suicide, and does the sex difference persist across age groups?

7) Planned plots for next assignment...

For Q1 (sex/race disparity over time)

Planned plots: - Line chart: YEAR (x) vs ESTIMATE (y), color = subgroup label (STUB_LABEL), filtered to a specific stratification scheme (STUB_NAME) such as sex or race/ethnicity. - Optional: Gap-over-time plot (difference between two groups per year). (longitudinal maybe?)

For Q2 (age groups most affected and change over time)

Planned plots: - Heatmap: YEAR (x) by AGE (y) with fill = ESTIMATE to detect shifting risk across decades. - Small-multiples line plots: ESTIMATE over time for selected age groups.

For Q3 (sex differences across age groups)

Planned plots: - Facet line chart: YEAR vs ESTIMATE, color by sex subgroup, faceted by AGE (or a subset of age bands). - Optional: Boxplots of ESTIMATE by sex within chosen age ranges/time windows. (boxplots are great to see the IQR and visualize outliers if needed to construct an argument.)

PS: Thank you for reading - if you have ANY questions or concerns (if my writing is not professional enough, need to reword certain sections, etc.) please contact me via email. I really care about this class and I want to get a good grade to show to my mentor that taught me R at RAND. :) I worked on a suicide research project across Pittsburgh collecting data so this is very important to me.